

Econometrics Preliminary Exam
Agricultural and Resource Economics, UC Davis

June 26, 2024

There are **FOUR** questions. Answer each part of each question. All questions are weighted equally. Within each question, each part will receive equal weight in grading. You have 20 minutes to read the exam and then four hours to complete the exam.

I. Probability and Statistics

- (a) Consider two random variables X and Z that are independent of each other. Define $W = X + Z$.
- Suppose both X and Z are binary. Suppose $\mathbb{P}(X = 1) = p_x \in (0, 1)$ and $\mathbb{P}(Z = 1) = p_z \in (0, 1)$. Find the probability mass function of W .
 - Following the previous question, find $\mathbb{P}(X = 1|W = 1)$, $\mathbb{P}(Z = 0|W = 1)$, and $\mathbb{P}(X = 1, Z = 0|W = 1)$. Is X independent of Z after conditioning on $W = 1$?
 - When does the condition $\mathbb{E}[X \cdot Z|W] = \mathbb{E}[X|W] \cdot \mathbb{E}[Z|W]$ hold? (Disregard the X, Z distribution given in part i.)
- (b) Consider an i.i.d. sample of $\{Z_i\}_{i=1}^n$ following the exponential distribution. Its probability density function (pdf) is $f_Z(z) = \lambda e^{-\lambda z} \cdot 1(z \geq 0)$ with a constant $\lambda > 0$. Define $\theta = 1/\lambda$. $\mathbb{E}[Z_i] = \theta$ and $\mathbb{V}[Z_i] = \theta^2$.
- Propose a moment estimator for θ . Call the estimator $\hat{\theta}$. Show consistency of $\hat{\theta}$. Derive the asymptotic distribution of $\sqrt{n}(\hat{\theta} - \theta)$ as $n \rightarrow \infty$.
 - Propose a consistent estimator for λ . Call the estimator $\hat{\lambda}$. Show consistency of $\hat{\lambda}$. Derive the asymptotic distribution of $\sqrt{n}(\hat{\lambda} - \lambda)$ as $n \rightarrow \infty$.
- (c) Consider random variables Z and A and finite constants m and v . Note that if $Z|A \sim N(A, 1)$ and $A \sim N(m, v)$, then $Z \sim N(m, v + 1)$. Also note that the pdf of a random variable $X \sim N(\mu, \sigma^2)$ is $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.
- What is the marginal pdf of Z ? What is the joint pdf of Z and A ?
- Now, consider an i.i.d. sample of $\{(Z_i, A_i)\}_{i=1}^n$ such that $Z_i|A_i \sim N(A_i, 1)$ and $A_i \sim N(0, v)$. (Please note that the mean of A_i is simplified to zero.) The sample of $\{Z_i\}_{i=1}^n$ is observed, while the sample of $\{A_i\}_{i=1}^n$ is unobserved.
- Derive the maximum likelihood estimator (MLE) of v .
 - Is your MLE estimator unbiased? Why or why not?

II. Linear Regression

Consider the model $Y_i = X_i' \beta + e_i$, where X_i is a $k \times 1$ vector and $E[X_i e_i] = 0$. You have an *iid* random sample of size n . Consider the following estimators:

$$\hat{\beta} = \left(\sum_{i=1}^n X_i X_i' \right)^{-1} \left(\sum_{i=1}^n X_i Y_i \right)$$
$$\bar{\beta} = \left(\sum_{i=1}^n w_i X_i X_i' \right)^{-1} \left(\sum_{i=1}^n w_i X_i Y_i \right),$$

where w_i is a scalar.

- (a) Is $\hat{\beta}$ consistent for β ? If so, prove it. If not, state additional conditions you require for consistency and prove consistency under those conditions.
- (b) Is $\bar{\beta}$ unbiased for β ? If so, prove it. If not, either (i) state additional conditions you require for unbiasedness and prove unbiasedness under those conditions, or (ii) explain why no such conditions exist.
- (c) Is $\bar{\beta}$ consistent for β ? If so, prove it. If not, state additional conditions you require for consistency and prove consistency under those conditions.
- (d) Following on from (c), find the asymptotic distribution of $\sqrt{n}(\bar{\beta} - \beta)$ as $n \rightarrow \infty$. You may carry over any assumptions you made in (c) to show consistency. State any additional assumptions you require.
- (e) Assume $w_i = 1(Y_i > \bar{Y}_i)$, i.e., w_i equals one if Y_i exceeds its mean and zero otherwise. Is $\bar{\beta}$ consistent for β ? If so, prove it. If not, either (i) state additional conditions you require for consistency and prove consistency under those conditions, or (ii) explain why no such conditions exist.
- (f) Assume w_i is a function of X_i , $E(e_i^2 | X_i) = \sigma^2 w_i^{-1}$ and $E(e_i | X_i) = 0$. Which of $\bar{\beta}$ and $\hat{\beta}$ is a more efficient estimator for β ? Justify your answer with a proof.
- (g) Assume w_i is a function of X_i and $E(e_i^2 | X_i) = \sigma^2 w_i^{-1}$. Describe how you would construct a bootstrap confidence interval for β . Be specific, including about which estimator you would use and why.

III. Models with Heterogeneous Coefficients with Cross-sectional and Panel Data

- (a) Consider the model $Y_i = X_i\beta_i + e_i$, where X_i is a scalar and $E(X_i e_i) = 0$. The random coefficient is defined as $\beta_i = \gamma_0 + \gamma_1 X_i$, where γ_0 and γ_1 are unknown parameters.
- Derive the probability limit of the OLS estimator $\hat{\beta} = \frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2}$. State any additional assumptions you make.
 - Propose a test of the null hypothesis that β_i is constant across i .
 - Derive the asymptotic null distribution of the test statistic you proposed in III(a)ii. State any additional assumptions you make.
- (b) Consider the linear panel data model, where for $i = 1, \dots, n$, $t = 1, \dots, T$

$$Y_{it} = X'_{it}\beta_0 + a_i + u_{it},$$

where $k = \dim(X_{it}) = \dim(\beta_0)$. All asymptotic statements in this question pertain to $n \rightarrow \infty$, holding T fixed.

- Assuming the sequential exogeneity of X_{it} , propose a consistent estimator of β_0 that relies on an over-identifying set of moment conditions. If the estimator admits a closed-form solution, solve for it.
- Provide sufficient conditions for the consistency of the estimator you propose in III(b)i.
- Now suppose that the true outcome model is given by the following

$$Y_{it} = X'_{it}\beta_i + a_i + u_{it}.$$

Derive the probability limit of the estimator you propose in III(b)i under this outcome model, maintaining the sequential exogeneity of X_{it} . Prove the result and state any conditions required for it.

IV. Nonlinear Estimation with Cross-sectional and Panel Data

Given a dataset of $\{Y_i, X_i\}_{i=1}^n$, consider the nonlinear model,

$$Y_i = m(X_i, \theta_0) + e_i,$$

where θ_0 is a k -dimensional parameter. All asymptotic statements in this question pertain to $n \rightarrow \infty$.

- (a) Define the nonlinear least squares estimator, $\hat{\theta}$, and its population analogue. Provide primitive sufficient conditions for the uniqueness identification of θ_0 . Prove that these conditions are sufficient.
- (b) Provide sufficient conditions for the asymptotic normality of $\sqrt{n}(\hat{\theta} - \theta_0)$. You can provide high-level conditions, but make sure to derive the mean value expansion required for the result and invoke the appropriate theorems.
- (c) Propose an estimator of the asymptotic variance of $\sqrt{n}(\hat{\theta} - \theta_0)$ and provide high-level conditions for its consistency. Prove that the conditions you provide are sufficient.
- (d) Write down the Wald statistic to test $H_0 : c(\theta_0) = 0$, where $c(\theta)$ is a q dimensional, known function that is differentiable in θ . Provide a proof sketch of its asymptotic distribution under H_0 . State any conditions you require for the result.

Note: In your answer, you might find the following theorem from Hansen (2022) *Probability and Statistics for Economists* helpful. Let X be an $r \times 1$ random variable. *Theorem 5.19: If $X \sim N(0, A)$ with an $r \times r$ positive definite A , then $X' A^{-1} X \sim \chi_r^2$.*

- (e) Now suppose that you observe panel data of Y_{it} and X_{it} for $i = 1, \dots, n$, $t = 1, 2$. Furthermore, suppose that the outcome is given by the following,

$$Y_{it} = \mu(X_{it}, \gamma_0) + a_i + u_{it}.$$

- i. Propose an estimator of γ_0 that would be consistent regardless of the dependence between a_i and X_{it} .
- ii. Provide primitive sufficient conditions for the uniqueness of the population analogue of the estimator you propose. Prove that they are sufficient.
- iii. Comment briefly on how you would provide conditions for the asymptotic normality of this estimator.